

Dice Article Study

Polymer state formation in solvent based on Monte Carlo Simulation

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Concept

- **Ergodicity**: the nature of experiencing all possible conditions after sufficient time
- **Barrier of rotation**: the energy that molecules must overcome as they change shape
- **To check the various states of the molecules**, ergodicity must be secured using algorithms that can be sampled evenly without being trapped by a high barrier of rotation.
- Application of the method of cutting molecules and assembling them back to the lower energy side

Concept

- S-QM/MM
- QM: a precise quantum mechanical calculation; the important part is calculated like this (solute)
- MM: Fast molecular calculations; the surrounding extra molecules are calculated like this (solvent)
- S-: Sequence calculation; after MM calculation, the critical part is executed by QM calculation

Abstract & Introduction

- **Limitations of MD:** High 'energy barrier' to changes in the state of molecules, trapped in only one shape
- **Advantages of MC:** On the other hand, the Monte Carlo (MC) method is advantageous for identifying various molecular structures

Metropolis MC

- Step 1: Initial state formation
- Step 2: Random state change
- Step 3: Check the potential change
- Step 4: repetition

Probability from state "old" to "new"

Probability of formation from "old" to "new"

$$\mathcal{K}(o \rightarrow n) = P(o)\alpha(o \rightarrow n)\text{acc}(o \rightarrow n)$$

Probability of state "old"

Probability of acceptance

Bias Monte Carlo Sampling

- Step 1: Initial state formation

- **Step 2: Random state change**

- Step 3: Check the potential change

- Step 4: repetition

$$\mathcal{U}(n) = \mathcal{U}^{\text{hard}}(n) + \mathcal{U}^{\text{soft}}(n)$$

Enough not to change
(bond distance, bond
angle..)

Enough to change
(rotation angle, dihedral
angle)

Bias Monte Carlo Sampling

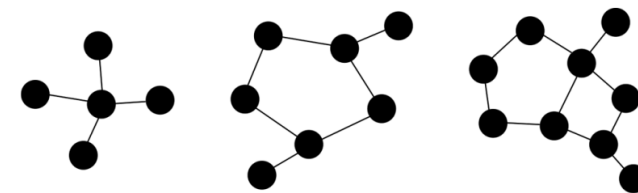
$$\mathcal{U}^{\text{hard}}(n) = \sum_{i=1}^{N_{\text{frag}}} u_i(n)$$

Dividing the molecule into N fragments, Potential of hard is defined as the sum of energy of each fragment.

$$\mathcal{U}^{\text{soft}}(n) = \sum_{i=2}^{N_{\text{frag}}} \sum_{j=1}^{i-1} u_{ij}(n) + \sum_{i=1}^{N_{\text{frag}}} \sum_{\substack{X=1 \\ X \neq A}}^N u_{i-X}(n) = \sum_{i=1}^{N_{\text{frag}}} u_{i\alpha}(n)$$

Interaction with connected fragments

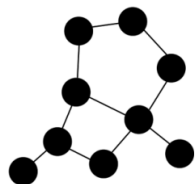
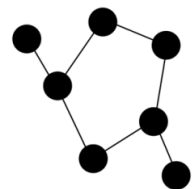
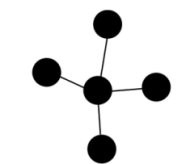
Interaction with disconnected, solvent fragments



Set non-critical parts to fragments
Less critical parts) -> Increasing
computational efficiency

Fragment setting

- R; Rigid; Fix the shape
- F: semiflexible; Fix bond distance
- M; flexible; unfix bond distance, unfix bond angle

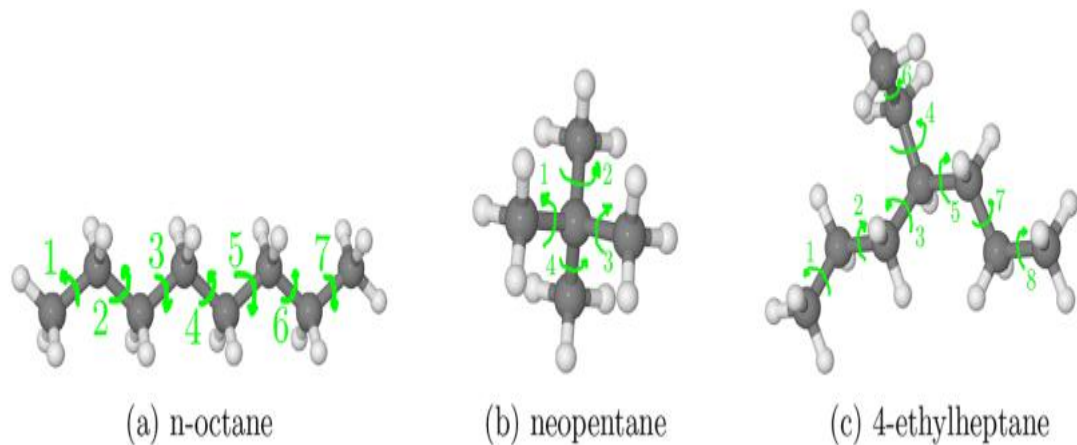


- Fragment type
- fragments with a central atom: rotation with central atoms
- simple ring: rotation with ring
- generic fragments: no constraint

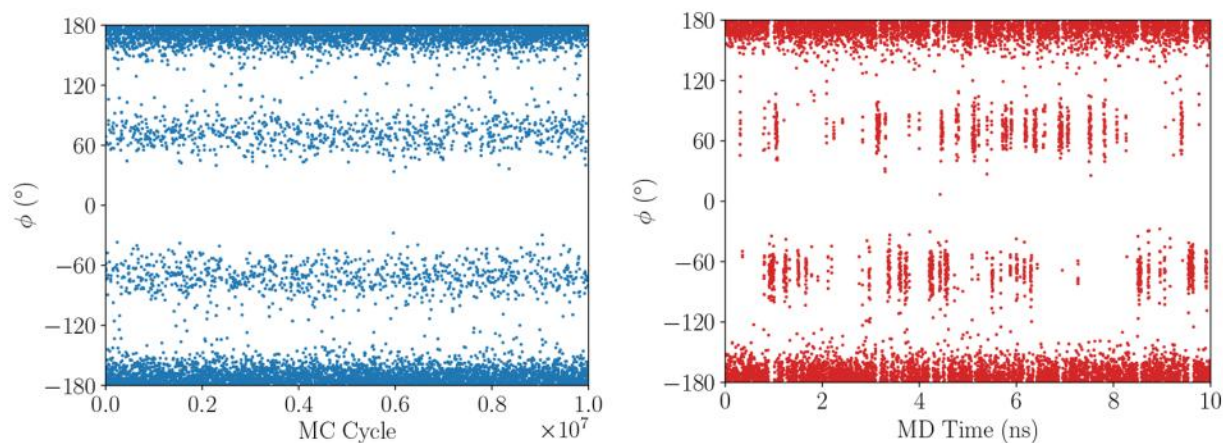
Soft setting

- Step 1: Molecular reconstruction (o)
 - Run reconfiguration for polymer fragments
- Step 2: Probability calculation
 - When each fragment is attached at a specific angle, calculate the potential
 - Repeat until all fragments are stuck
- Step 3: Determination
 - Based on the potential comparison, determine configuration

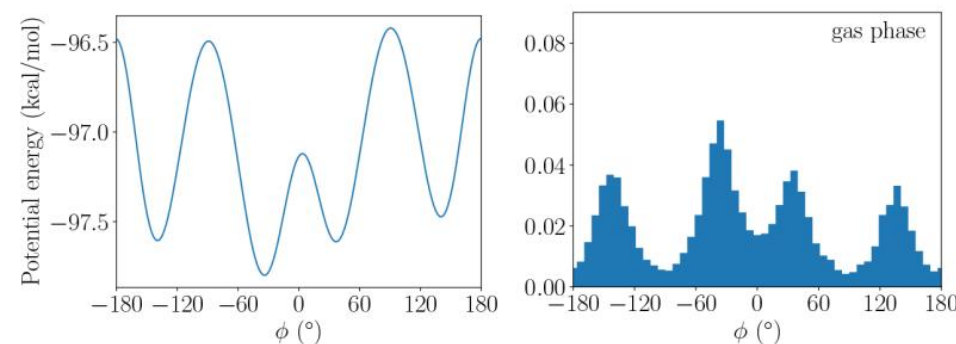
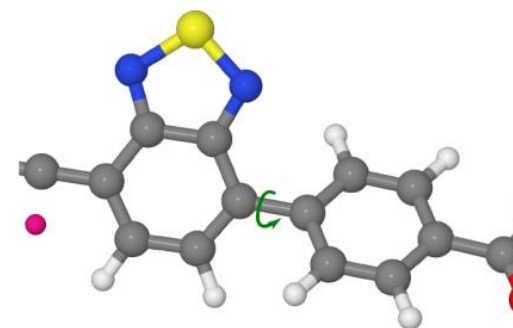
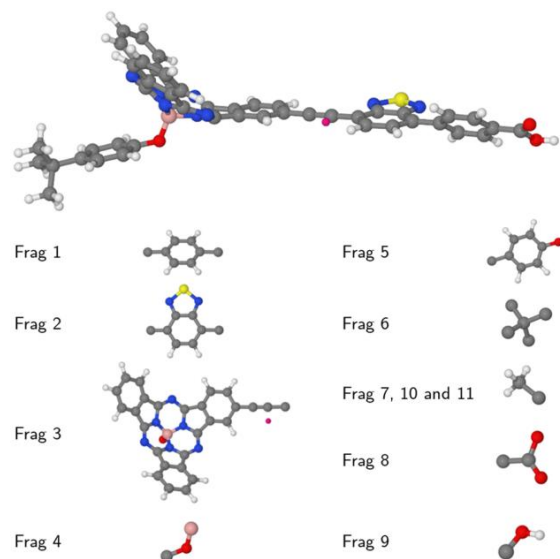
Result 1; Alkanes dihedral angle



- Dihedral angle from MD and that from Dice is similar



Result 2; SubPC



The method of cutting the polymer into several fragments and reattaching them in the water

-> Check what you can see in the water

Potential can be checked as the angle changes

Conclusion

- By dividing polymer properties into changing and unchanging parts, the MC process can be accelerated.
- By dividing the polymer into fragments and reassembling it, the process of high energy barriers to state change can be well expressed.